

PROGRAMMING

RaceDay

API Endpoint Plan

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1. Authentication

Method	Route	Description	Role	Request Body	Response
POST	/api/auth/register	Register a new Organiser or Participant. Based on the role field, the record is created in the Organisers or Participants table.	Public	{ email, password, fullName, role, organisationName?, contactPhone?, dateOfBirth?, username? }	201 Created – user created 400 Bad Request – validation errors 409 Conflict – email already exists
POST	/api/auth/login	Authenticate against either the Organisers or Participants table and receive a JWT token carrying the user's role.	Public	{ email, password }	200 OK – returns token and user details 401 Unauthorized – invalid credentials

2. User Profile

Method	Route	Description	Role	Request Body	Response
GET	/api/users/me	Retrieve the current user's profile. The API reads the role from the JWT and queries Organisers or Participants accordingly, returning the matching shape of data.	Authenticated	None	200 OK – organiser or participant profile data 404 Not Found – profile missing
PUT	/api/users/me	Update the current user's own profile. The API reads the role from the JWT and updates the matching table.	Authenticated	Organiser: { fullName, organisationName?, contactPhone? } Participant: { fullName, dateOfBirth?, username? }	200 OK – updated profile 400 Bad Request – validation errors

3. Events

Method	Route	Description	Role	Request Body	Response
GET	/api/events	List all events, with optional filters (date, type, location).	Authenticated	None	200 OK – array of events
POST	/api/events	Create a new event, linked to the logged-in organiser.	Organiser	{ eventName, eventDescription, eventDate, eventLocation, distance, eventType }	201 Created – event details 400 Bad Request – invalid data
GET	/api/events/{id}	Get details of a specific event, including its categories.	Authenticated	None	200 OK – event with categories 404 Not Found – event does not exist

Method	Route	Description	Role	Request Body	Response
PUT	/api/events/{id}	Update an existing event.	Organiser	{ eventName, eventDescription, eventDate, eventLocation, distance, eventType }	200 OK – updated event 403 Forbidden – not the organiser who owns this event 404 Not Found
DELETE	/api/events/{id}	Delete an event (cascades categories, enrolments, results).	Organiser	None	204 No Content – deleted 403 Forbidden – not the organiser who owns this event 404 Not Found
GET	/api/events/upcoming	List upcoming events (event_date in the future).	Authenticated	None	200 OK – array of upcoming events

4. Categories

Method	Route	Description	Role	Request Body	Response
GET	/api/events/{eventId}/categories	List all categories for a specific event.	Authenticated	None	200 OK – array of categories 404 Not Found – event missing
POST	/api/events/{eventId}/categories	Create a new category for an event.	Organiser	{ categoryName, categoryDescription }	201 Created – category details 404 Not Found – event missing
PUT	/api/categories/{id}	Update a category.	Organiser	{ categoryName, categoryDescription }	200 OK – updated category 403 Forbidden – not organiser of the parent event 404 Not Found
DELETE	/api/categories/{id}	Delete a category.	Organiser	None	204 No Content 403 Forbidden 404 Not Found

5. Event Enrolments

Method	Route	Description	Role	Request Body	Response
POST	/api/events/{eventId}/enrol	Participant enrolls in an event by selecting a category. Uniqueness of one enrolment per participant per event is enforced at the application level, since Enrolments links to Categories rather than directly to Events.	Participant	{ categoryId }	201 Created – enrolment record 400 Bad Request – already enrolled in this event, or category invalid 404 Not Found – event/category missing
GET	/api/events/{eventId}/enrolments	View all enrolments for an event, with participant details.	Organiser	None	200 OK – list of enrolments 403 Forbidden – not organiser of event 404 Not Found
GET	/api/users/me/enrolments	View the current participant's own enrolments.	Participant	None	200 OK – list of enrolments, with event and category details
GET	/api/events/{eventId}/enrolments/export	Export enrolments for an event as CSV.	Organiser	None	200 OK – CSV file download 403 Forbidden – not organiser of event 404 Not Found

6. Results

Method	Route	Description	Role	Request Body	Response
PUT	/api/enrolments/{enrolmentId}/result	Record or update a participant's result. updated_by is set from the logged-in organiser's ID.	Organiser	{ finishTime, position, notes }	200 OK – result details 403 Forbidden – not organiser of the event this enrolment belongs to 404 Not Found – enrolment missing
GET	/api/users/me/results	View the current participant's own results, with event details.	Participant	None	200 OK – list of results

Notes

- eventType accepts Run, Walk, or Cycle, per the functional requirements.
- distance is the event's overall distance in kilometres (e.g. 42.2 for a marathon); individual categories may offer shorter distances within the same event.
- This plan will be implemented as-is in Part 2. Any deviation from this plan during implementation will be documented and explained in the Part 2 README.
- Failure responses shown are the primary expected ones; standard validation (400) and auth (401) failures apply across all authenticated/role-restricted routes.
- Because Enrolments does not carry a direct event_id, enforcing “one enrolment per participant per event” (rather than per category) needs an application-level check in Part 2 — see the SQL script's comments for the corresponding trade-off on the database side.